

DNV-RP-C205 · Strouhal · DNV-RP-C203

Vortex-induced vibration — screening, frequency & fatigue

A first-pass VIV toolkit for tubular members and risers: natural-frequency calculation (Euler-Bernoulli beam + tension-controlled string, five boundary conditions, fluid added mass), vortex-shedding frequency from the Strouhal relation, reduced-velocity lock-in screening with a safety factor, and VIV-induced fatigue on DNV-RP-C203 S-N curves — the full-member generalization of the lock-in check the CFD Re=100 cylinder card exposes. Backed by unit, CLI and integration tests and a standalone CI workflow.

4–8

cross-flow lock-in V_r band

13

DNV-RP-C203 S-N curves

5

beam boundary conditions

WHAT YOU GET

- Reduced-velocity lock-in screening live: $V_r = V/(f_n \cdot D)$ against the 4–8 cross-flow band with safe / marginal / lock-in status and safety factor
- Natural frequency from Euler-Bernoulli beam + tensioned-string models, five boundary conditions, fluid added mass
- Strouhal shedding frequency with Reynolds- and surface-condition-based selection (smooth / rough / straked / helical), plus current-profile sweeps
- VIV amplitude (A/D) to bending stress range to Palmgren-Miner damage and fatigue life across 13 tabulated S-N curves
- Config-driven tubular-member workflow emitting CSV result tables, runnable end-to-end from the example input.yml